Homework_06

March 2, 2026

Probability for Data Science

UC Berkeley, Spring 2026

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1 Homework 6 (Due March 2nd at 5 PM)

1.0.1 Instructions

Questions 1 and 2 can be answered based on Tuesday's lecture (and past content) alone. Your homeworks will generally have two components: a written portion and a portion that also involves code. Written work should be completed on paper, and coding questions should be done in the notebook. Start the work for the written portions of each section on a new page. You are welcome to LaTeX your answers to the written portions, but staff will not be able to assist you with LaTeX related issues.

It is your responsibility to ensure that both components of the homework are submitted completely and properly to Gradescope. **Make sure to assign each page of your pdf to the correct question. Refer to the bottom of the notebook for submission instructions.**

1.0.2 How to Do Your Homework

The point of homework is for you to try your hand at using what you've learned in class. The steps to follow:

- Go to lecture and sections, and also go over the relevant text sections before starting on the homework. This will remind you what was covered in class, and the text will typically contain examples not covered in lecture. The weekly Study Guide will list what you should read.
- Work on some of the practice problems before starting on the homework.
- Attempt the homework problems by yourself with the text, section work, and practice materials all at hand. Sometimes the week's lab will help as well. The two steps above will help this step go faster and be more fruitful.
- At this point, seek help if you need it. Don't ask how to do the problem — ask how to get started, or for a nudge to get you past where you are stuck. Always say what you have already tried. That helps us help you more effectively.

- For a good measure of your understanding, keep track of the fraction of the homework you can do by yourself or with minimal help. It's a better measure than your homework score, and only you can measure it.

1.0.3 Rules for Homework

- Every answer should contain a calculation or reasoning. For example, a calculation such as $(1/3)(0.8) + (2/3)(0.7)$ or `sum([(1/3)*0.8, (2/3)*0.7])`is fine without further explanation or simplification. If we want you to simplify, we'll ask you to. But just $\binom{5}{2}$ by itself is not fine; write "we want any 2 out of the 5 frogs and they can appear in any order" or whatever reasoning you used. Reasoning can be brief and abbreviated, e.g. "product rule" or "not mutually exclusive."
- You may consult others (see "How to Do Your Homework" above) but you must write up your own answers using your own words, notation, and sequence of steps.
- We'll be using Gradescope. You must submit the homework according to the instructions at the end of homework set.

1.1 We will not grade assignments which do not have pages selected for each question.

1. "Move to Front" Permutations

Consider an alphabet of length N. A move-to-front permutation of the N letters consists of picking one of the letters (randomly or otherwise) and moving it to the front of the list.

For example, if the alphabet consists of the three letters A, B, and C, and you start with the permutation ABC, then CAB is the result of one move-to-front (the chosen letter is C), as is ABC (the chosen letter is A), but not CBA.

a) As a preliminary, recall that all transition matrices are stochastic; that is, each row sums to 1. Suppose the transition matrix of a finite-state, irreducible, aperiodic chain is doubly stochastic; that is, each of its columns also sums to 1. Explain why the stationary distribution must be uniform.

A finite state, irreducible, aperiodic chain has a transition matrix that has the property of $\pi\mathbb{P} = \pi$, or it fulfills the balance equations.

If the transition matrix of this chain is doubly stochastic, that means that each column is also a probability distribution, as it all sums to 1.

In a row of a transition matrix, each element of the row means the probability of ending up in state j when starting in state i.

In the stationary distribution, it can also be interpreted as a particle spending j proportion of the time in state j, because the Markov property also tells us for a finite state, irreducible, aperiodic chain does not matter on where you start.

That means that each row of the n step transition matrix as $n \to \infty$ should be the same, π .

A transition matrix $\mathbb{P}$ is n x n, since there are n states.

If each column in the transition matrix sums up to 1, then that means that we spend the same amount of time in each state, which then means that each element of each row is the same. Thus, π is a uniform distribution, meaning that every element in the matrix is also the same.

Mathematically, a stationary distribution satisfies the balance equations:

$$\pi_j = \sum_{\forall i} \pi_i P_{ij} \quad \text{and} \quad \sum_{\forall i}^{\pi_i} = 1$$

If we assume the distribution is uniform ($\pi_i = \frac{1}{n} \forall i$) and the matrix is doubly stochastic

$$\pi_j = \sum_{i=1}^{n} \pi_i P_{ij} = \frac{1}{n} \sum_{i=1}^{n} P_{ij} = \frac{1}{n} \cdot 1 = \frac{1}{n}$$

Thus, $\pi_j = \frac{1}{n}$ is a valid solution to the balance equations.

b) A standard deck consists of 52 cards. A "random to front" shuffle is defined as follows: Pick one of the 52 cards uniformly at random and move it to the front of the deck (which you are welcome to think of as the top of the deck, if you prefer). Explain why if you perform this move over and over again, in the long run the deck will become well shuffled; that is, all permutations will be equally likely. [Hint: Set up an appropriate chain and use Part **a**. You might want to try it out first with an alphabet of just the letters A, B, and C.]

In a alphabet with ABC:

From ABC -> ABC, BAC, CAB

From ACB -> ACB, CAB, BAC

From BAC -> BAC, ABC, CBA

From BCA -> BCA, CBA, ABC

From CAB -> CAB, ACB, BCA

From CBA -> CBA, BCA, ACB

This chain is finite, irreducible, and aperiodic. In this chain, each valid random to front shuffle occurs with probability $\frac{1}{N}$, where N is the number of letters in the alphabet, or the number of cards in the deck.

There will also be N incoming and outgoing arcs from each node, since there are N letters to pick from each state, and N ways to get to each state. Therefore, since each probability is $\frac{1}{N}$, the transition matrix will be doubly stochastic.

For example, the transition matrix of the above MKC is:

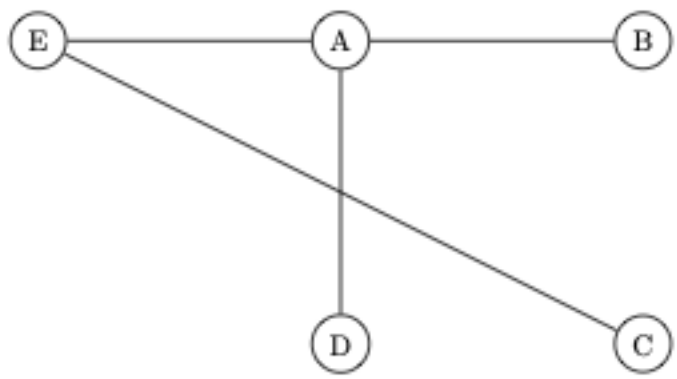
$$\mathbb{P} = \begin{bmatrix} ABC & ACB & BAC & BCA & CAB & CBA \\ \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & \frac{1}{3} \\ \frac{1}{3} & 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} \end{bmatrix}$$

In part a, we showed that in the long run, a doubly stochastic transition matrix will be the uniform distribution, with $\pi_i = \frac{1}{\#\text{ of states}} = \frac{1}{N!}$. Thus, if we perform the random to front shuffle over the long run, then it will result in a uniform distribution in drawing the cards.

2. Social Network

A social network consists of a finite set of members. Members i and j are said to be *linked* if either they are friends or they are *friends of friends* which is shown in an example and then formally defined below.

As an example, here is a social network that has five members A, B, C, D, and E. Friends are any two members that are connected by one line segment. For example, A and B are friends, and C and E are friends. Member A has 3 friends; B, C and D have one friend each; and E has two friends. Members B and C aren't friends but they are linked by the sequence of friends BAEC.



Formally, Members i and j are linked if either they are friends or for some $n \geq 1$ there are members $a_1, a_2, \ldots, a_n$ such that Members i and a_1 are friends, Members a_1 and a_2 are friends, Members a_2 and a_3 are friends, and so on, and Members a_n and j are friends.

Consider a network of m members. For $1 \leq i \leq m$ let f_i be the number of friends of Member i, and assume that $f_a \neq f_b$ for some members a and b. Also assume that every member is linked to every other member.

Let $X_0, X_1, X_2, \ldots$ be a Markov Chain on the set of members, with transitions based on the following proposal scheme. Given that the chain is at Member i:

- Select a friend uniformly at random from all of Member i's friends.
- If the selected friend is Member j, then:
  - If $f_j \leq f_i$, move to j.
  - If $f_j > f_i$, toss a coin that lands heads with chance f_i/f_j. If it lands heads, move to j. If it lands tails, stay at i.

a) For states $i \neq j$, find the one-step transition probability $P(i,j) = P(X_1 = j \mid X_0 = i)$.

$$
\begin{aligned}
P(i,j) &= P(X_1 = j \mid X_0 = i) \\
&= \begin{cases} 1 \cdot \frac{1}{f_i}, & \text{if } i \text{ and } j \text{ are friends and } f_j \leq f_i \\ \frac{f_i}{f_j} \cdot \frac{1}{f_i}, & \text{if } i \text{ and } j \text{ are friends and } f_j > f_i \\ 0, & \text{if } i \text{ and } j \text{ are not friends} \end{cases} \\
&= \begin{cases} \frac{1}{f_i} & \text{if } i \text{ and } j \text{ are friends and } f_j \leq f_i \\ \frac{1}{f_j} & \text{if } i \text{ and } j \text{ are friends and } f_j > f_i \\ 0, & \text{otherwise} \end{cases}
\end{aligned}
$$

b) Briefly explain why the chain has a steady state distribution, and find that distribution. Identify it as one of the famous ones and provide its name and parameters.

The transition matrix is symmetric by construction, because if i has more friends and is linked to j , then we propose a transition with probability $\frac{1}{f_i}$, while if j has more friends and is linked to i, we propose a transition with probability $\frac{1}{f_j}$. Otherwise, $P(i,j) = 0$. Thus, the denominator is just the member that has more friends, or $\max(f_i, f_j)$. Since $\max(f_i, f_j) = \max(f_j, f_i)$.

This transition matrix is also irreducible, as all Members are linked to each other (e.g BAE-CEADAB).

Since we know the matrix is symmetric, and that all rows sum up to 1 due to the definition of a Markov chain, we know that the matrix is also doubly stochastic. This is because in a symmetric matrix, row i = column j if $i = j$. We proved the previous question that a doubly stochastic transition matrix in the long run follows a uniform distribution.

Thus, the X_k has a uniform($\frac{1}{n}$) distribution.

1.2 3. Bounds

A random variable X, not necessarily non-negative, has $E(X) = 20$ and $SD(X) = 4$. In each part below **find the best bounds you can** based on the information given.

a) Find upper and lower bounds for $P(0 < X < 40)$.

Using Chebysev's Inequality:

$$\begin{aligned} P(0 < X < 40) \leq 1 - P(|X - \mu_X| \geq c) &= 1 - P(|X - \mu_X| \geq 20) \\ &= 1 - \frac{4^2}{20^2} \\ &= 1 - \frac{1}{25} \\ &= \frac{24}{25} \end{aligned}$$

Thus,

$$\boxed{\frac{24}{25} \leq P(0 < X < 40) \leq 1}$$

b) Find upper and lower bounds for $P(10 < X < 40)$.

We still use Chebysev's inequality, however it is two tailed. We cannot get a one tailed version. Thus, to get a lower bound, we need to look at a symmetric bound of $P(10 < X < 30) \leq P(10 < X < 40)$

$$\begin{aligned} P(10 < X < 30) &\geq 1 - P(|X - \mu_X| \geq c) \\ &= 1 - \frac{4^2}{10^2} \\ &= \frac{21}{25} \end{aligned}$$

Our best bounds is

$$\boxed{\frac{21}{25} \leq P(10 < X < 40) \leq 1}$$

c) Find an upper bound for $P(X \geq 40)$.

We cannot use Markov's Inequality because X is not nonnegative. The best bounds we could have is from Chebysev's.

Thus we need to find $P(X \geq 40 \cup X \leq 0) = 1 - P(0 < X < 40)$

From part a, we know $P(0 < X < 40) \leq \frac{24}{25}$

$$\boxed{0 \leq P(X \geq 40) \leq \frac{1}{25}}$$

d) Find an upper bound for $P(X^2 \geq 900)$.

For X^2, we can now guarantee that this is a nonnegative random variable, thus we can use Markov's Inequality.

$$P(X \geq c) \leq \frac{E[X]}{c}$$

$$P(X^2 \geq 900) \leq \frac{E[X^2]}{900}$$

$$\begin{aligned}
\text{Var}(X) &= E[X^2] - E[X]^2 \\
E[X^2] &= \text{Var}(X) + E[X]^2 \\
&= SD(X)^2 + E[X]^2 \\
&= 4^2 + 20^2 \\
&= 416
\end{aligned}$$

$$0 \leq P(X^2 \geq 900) \leq \frac{416}{900}$$

However, this might not be the best bound, since the squared term adds a lot of variance, and we can also partition the space to calculate Chebyshev's.

$$P(X^2 \geq 900) = P(|X| \geq 30) = P\{X \geq 30 \cup X \leq -30\}$$

Since $P\{X \geq 30 \cup X \leq -30\} = 1 - P(-30 < X < 30)$, we can bound this function in a similar way to part b.

Since $P(-30 < X < 30) \geq P(10 < X < 30)$ and $P(10 < X < 30) \geq \frac{21}{25}$ by part b.

$$\frac{21}{25} \leq P(10 < X < 30) \leq P(-30 < X < 30)$$

From this equation, we can assert that the upper bound for $P\{X \geq 30 \cup X \leq -30\} \leq \frac{4}{25}$

Thus the best bounds for $P(X^2 \geq 900)$ is $\boxed{0 \leq P(X^2 \geq 900) \leq \frac{4}{25}}$

4. Bias and Variance

Consider a probabilistic model that has a numerical parameter θ. A "probabilistic model" is just a set of assumptions about randomness. Let the random variable T be an estimator of θ. Frequently, T is a statistic based on a random sample.

The *bias* of T is defined as

$$B_\theta(T) \;=\; E_\theta(T) - \theta,$$

where the subscript θ reminds us that θ is the true value of the parameter. Note that $B_\theta(T)$ is a real-valued function of θ, not a random variable.

Recall that T is an unbiased estimator of θ if $E_\theta(T) = \theta$, that is, if $B_\theta(T) = 0$. But in general, estimators are biased.

The *mean squared error* of the estimator T is

$$MSE_\theta(T) \;=\; E_\theta\big((T-\theta)^2\big)$$

Follow the calculation in Section 12.2 of the textbook to show the *bias-variance decomposition* given by

$$MSE_\theta(T) \;=\; Var_\theta(T) + B_\theta^2(T)$$

Note that the square in the bias term makes sense. Bias has the same units as T, whereas the MSE and variance are in the square of those units.

$$\begin{aligned}
\mathrm{MSE}_\theta(T) &= E_\theta\left((T-\theta)^2\right) \\
&= E_\theta\left(T^2 - 2T\theta + \theta^2\right) \\
&= E_\theta(T^2) + E_\theta(-2T\theta) + E_\theta(\theta^2) \\
&= E_\theta(T^2) - 2\theta E_\theta(T) + \theta^2 \\
&= E_\theta(T^2) - E_\theta(T)^2 + E_\theta(T)^2 - 2\theta E_\theta(T) + \theta^2 \\
&= \mathrm{Var}_\theta(T) + \theta^2 - 2\theta E_\theta(T) + E_\theta(T)^2 \\
&= \mathrm{Var}_\theta(T) + (E_\theta(T) - \theta)^2 \\
&= \mathrm{Var}_\theta(T) + B_\theta^2(T)
\end{aligned}$$

Submission Instructions

Many assignments throughout the course will have a written portion and a code portion. Please follow the directions below to properly submit both portions.

1.2.1 Written Portion

- Scan all the pages into a PDF. You can use any scanner or a phone using applications such as CamScanner. Please **DO NOT** simply take pictures using your phone.
- Please start a new page for each question. If you have already written multiple questions on the same page, you can crop the image in CamScanner or fold your page over (the old-fashioned way). This helps expedite grading.
- It is your responsibility to check that all the work on all the scanned pages is legible.
- If you used LaTeX to do the written portions, you do not need to do any scanning; you can just download the whole notebook as a PDF via LaTeX.

1.2.2 Code Portion

- Save your notebook using `File > Save Notebook`.
- Generate a PDF file using `File > Save and Export Notebook As > PDF`. This might take a few seconds and will automatically download a PDF version of this notebook.
  - If you have issues, please post a follow-up on the general Homework 6 Ed thread.

1.2.3 Submitting

- Combine the PDFs from the written and code portions into one PDF. Here is a useful tool for doing so.
- Submit the assignment to Homework 6 on Gradescope.
- **Make sure to assign each page of your pdf to the correct question.**
- **It is your responsibility to verify that all of your work shows up in your final PDF submission.**

If you are having difficulties scanning, uploading, or submitting your work, please read the Ed Thread on this topic and post a follow-up on the general Homework 6 Ed thread.

1.3 We will not grade assignments which do not have pages selected for each question.

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